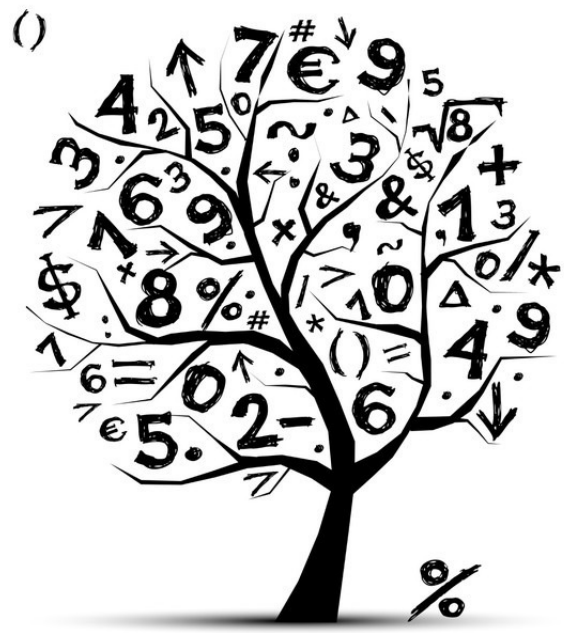


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Norms



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1 Introduction to norms

- A **norm** is a mathematical function that assigns a non-negative length or size to vectors. While there are many different types of norms, they all share three essential properties that ensure they behave consistently and meaningfully.

Definition: Norm.

A function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is called a **norm** if it satisfies the following three properties for all vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ and all scalars $\lambda \in \mathbb{R}$:

1. Non-negativity and definiteness:

$$\|\mathbf{p}\| \geq 0 \quad \text{and} \quad \|\mathbf{p}\| = 0 \iff \mathbf{p} = \mathbf{0}$$

The norm is always non-negative, and it equals zero if and only if the vector is the zero vector.

2. Absolute homogeneity (or scalability):

$$\|\lambda \mathbf{p}\| = |\lambda| \|\mathbf{p}\|$$

Scaling a vector by a scalar λ scales its norm by the absolute value $|\lambda|$.

3. Triangle inequality (or subadditivity):

$$\|\mathbf{p} + \mathbf{w}\| \leq \|\mathbf{p}\| + \|\mathbf{w}\|$$

The norm of the sum of two vectors is at most the sum of their norms.

- **Non-negativity and definiteness** give the norm a reliable meaning as “size” or “length.” They ensure that the norm is never negative and that the only vector with zero length is the zero vector.
- **Absolute homogeneity** ensures that scaling a vector scales its length proportionally. This matches intuition: stretching a vector makes it longer in the same proportion, while reversing its direction does not change its length.
- **Triangle inequality** captures the idea that taking a detour cannot be shorter than going directly. In other words, combining two displacements cannot produce a length larger than the sum of their lengths, reflecting the “straight line is shortest” intuition behind distance.
- Different norms may emphasize different aspects of a vector (e.g., its largest component, the sum of absolute values, or the geometric length), but they all respect these fundamental axioms.

2 ℓ_1 norm

Definition 1: ℓ_1 norm

The ℓ_1 **norm** (also called **Manhattan norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j| \quad (1)$$

- The ℓ_1 **norm** represents the sum of the absolute values of all components.
- In $\mathbb{R}^2$, the ℓ_1 norm corresponds to the **Manhattan distance** (the distance a taxi would travel on a grid-like street network).
- The ℓ_1 norm is widely used in **robust statistics** and **sparse optimization** (e.g., Lasso regression).

Example 1: ℓ_1 norm

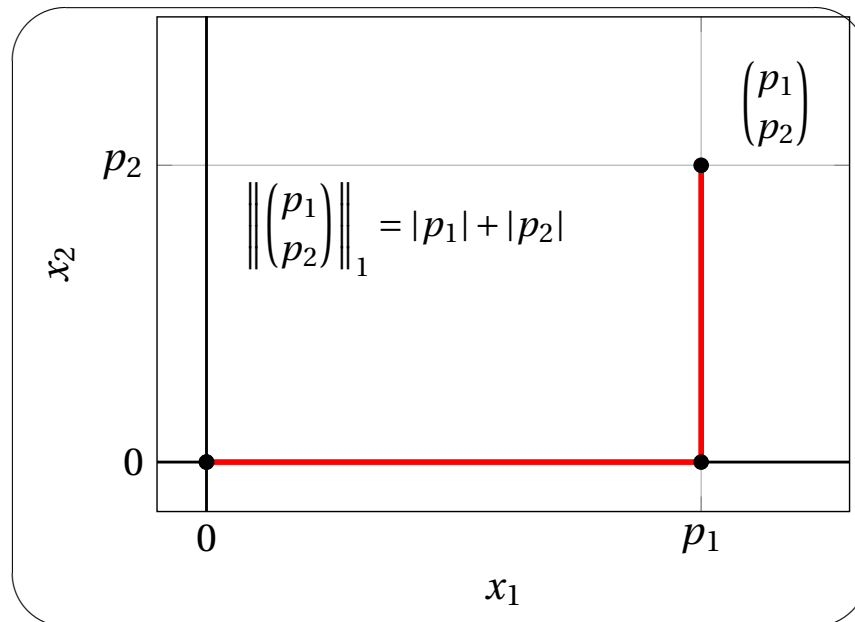
Consider the vectors:

$$\mathbf{p} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} \in \mathbb{R}^2, \quad \mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ -2 \\ 3 \end{pmatrix} \in \mathbb{R}^4$$

Their ℓ_1 norms are:

$$\begin{aligned} \left\| \begin{pmatrix} 3 \\ -4 \end{pmatrix} \right\|_1 &= |3| + |-4| = 3 + 4 = 7, \\ \left\| \begin{pmatrix} 1 \\ 2 \\ -2 \\ 3 \end{pmatrix} \right\|_1 &= |1| + |2| + |-2| + |3| = 1 + 2 + 2 + 3 = 8 \end{aligned}$$

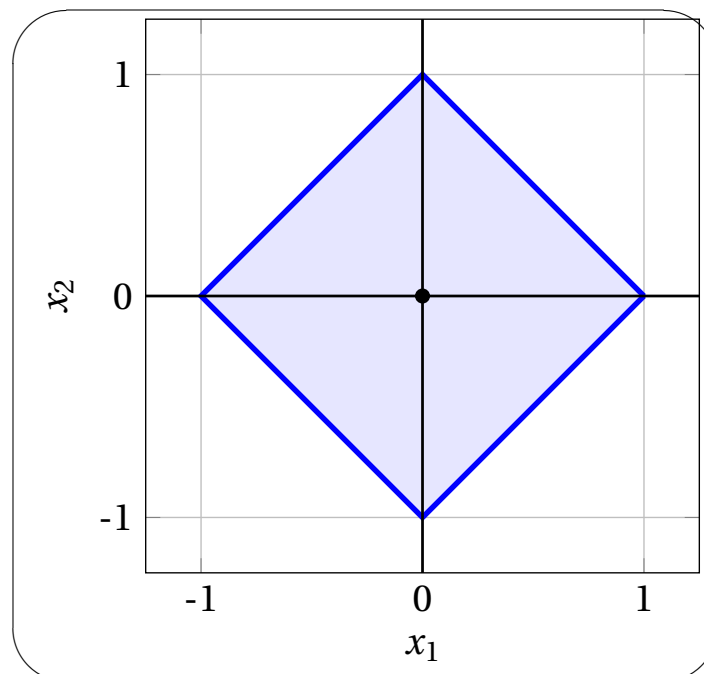
- The ℓ_1 **norm** of a column vector in $\mathbb{R}^n$ represents the **sum of the absolute values of its components**. In $\mathbb{R}^2$, this corresponds to the **Manhattan distance** (the distance traveled along grid lines), as illustrated by the following picture:



- In two dimensions, all the points at ℓ_1 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_1 \leq 1\}$$

are the points that lie on or inside a **diamond shape** (square rotated by 45°) centered at the origin.



2.1 Properties

- The ℓ_1 norm satisfies the three fundamental properties that define a norm.

Observation 1: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_1 \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_1 = 0 \iff \mathbf{p} = \mathbf{0} \quad (2)$$

Proof. From the definition $\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j|$.

1. **Non-negativity:** Since $|p_j| \geq 0$ for all $j \in \{1, 2, \dots, n\}$, and the sum of non-negative numbers is non-negative, we have

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j| \geq 0 \quad \text{for all } \mathbf{p} \in \mathbb{R}^n.$$

2. **Definiteness:**

($\Rightarrow$) If $\|\mathbf{p}\|_1 = 0$, then $\sum_{j=1}^n |p_j| = 0$. Since each $|p_j| \geq 0$, the sum equals zero only if $|p_j| = 0$ for all $j \in \{1, 2, \dots, n\}$, hence $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, i.e., $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, so

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |0| = 0.$$

□

Observation 2: Absolute homogeneity

For every scalar $\lambda \in \mathbb{R}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_1 = |\lambda| \|\mathbf{p}\|_1 \quad (3)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\lambda \mathbf{p}\|_1 &= \sum_{j=1}^n |\lambda p_j| \\ &= \sum_{j=1}^n |\lambda| |p_j| \quad (\text{property of absolute value}) \\ &= |\lambda| \sum_{j=1}^n |p_j| \\ &= |\lambda| \|\mathbf{p}\|_1 \end{aligned}$$

□

Observation 3: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_1 \leq \|\mathbf{p}\|_1 + \|\mathbf{w}\|_1 \quad (4)$$

Proof. For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} + \mathbf{w}\|_1 &= \sum_{j=1}^n |p_j + w_j| \\ &\leq \sum_{j=1}^n (|p_j| + |w_j|) \quad (\text{triangle inequality for absolute value}) \\ &= \sum_{j=1}^n |p_j| + \sum_{j=1}^n |w_j| \\ &= \|\mathbf{p}\|_1 + \|\mathbf{w}\|_1 \end{aligned}$$

□

3 ℓ_2 norm

Definition 2: ℓ_2 norm

The ℓ_2 **norm** (also called **Euclidean norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_2 = \underbrace{\sqrt{\sum_{j=1}^n p_j^2}}_{\sqrt{\mathbf{p}' \mathbf{p}}} \quad (5)$$

- When the subscript is omitted, i.e., when we write $\|\mathbf{p}\|$ instead of $\|\mathbf{p}\|_2$, the ℓ_2 norm is implied by default.

Example 2: ℓ_2 norm

- Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2,$$

Their norms are:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}, \quad \left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{4^2 + 1^2} = \sqrt{16 + 1} = \sqrt{17}$$

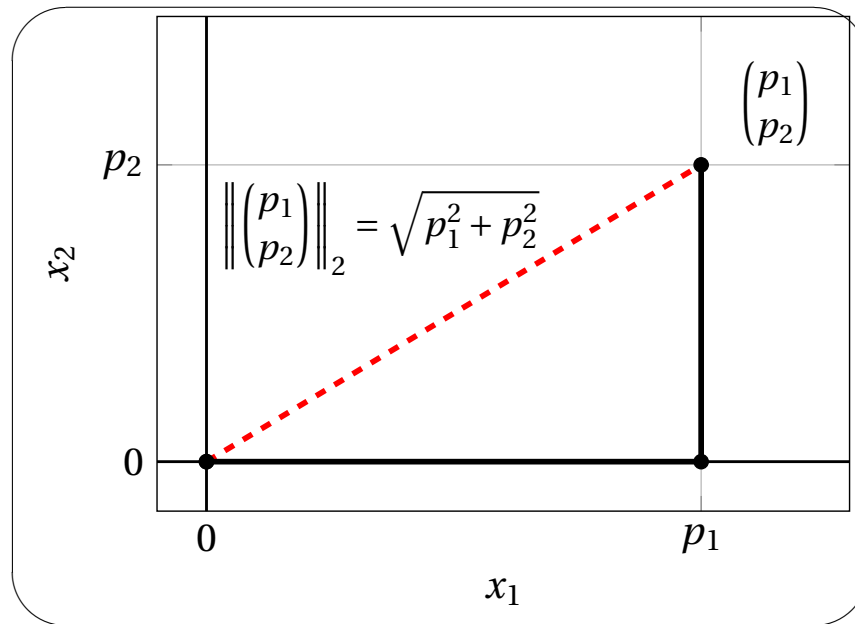
- Consider the four vertices of the diamond-shaped ℓ_1 unit ball in $\mathbb{R}^2$:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

Their norms are:

$$\left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 0^2} = 1, \quad \left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{0^2 + 1^2} = 1, \quad \left\| \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\|_2 = \sqrt{(-1)^2 + 0^2} = 1, \quad \left\| \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\|_2 = \sqrt{0^2 + (-1)^2} = 1$$

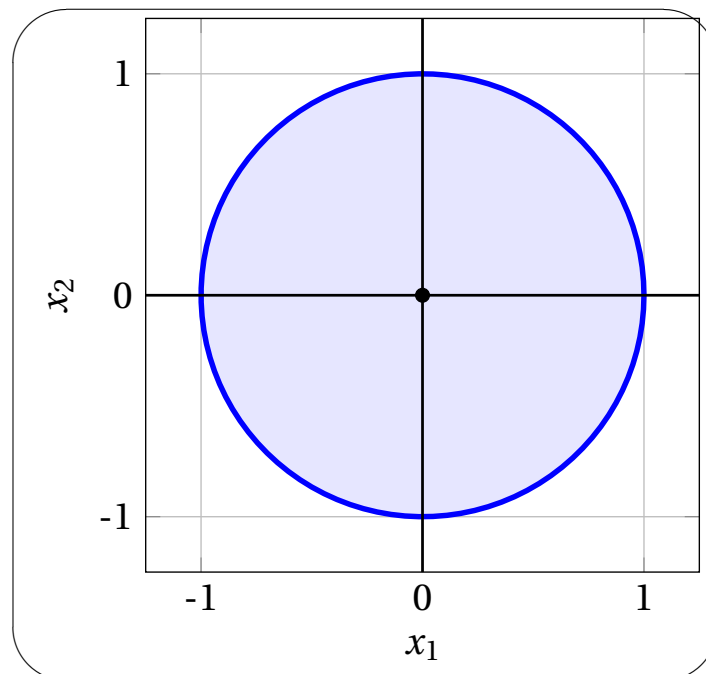
- The ℓ_2 **norm** of a column vector in $\mathbb{R}^n$ represents its **Euclidean distance from the origin**. This interpretation follows from the **Pythagorean theorem**, as illustrated in $\mathbb{R}^2$ by the following picture:



- In two dimensions, all the points at ℓ_2 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_2 \leq 1\}$$

are the points that lie on or inside the **unit circle** centered at the origin.



Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

the norm of the **difference** of the two vectors is:

$$\|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{\sum_{j=1}^n (p_j - w_j)^2}$$

This quantity corresponds both to the **Euclidean distance** between the two points $\mathbf{p}$ and $\mathbf{w}$, and to the **length of the diagonal** of the parallelogram spanned by those points, starting from one and pointing to the other.

Example 3: ℓ_2 norm the difference of vectors

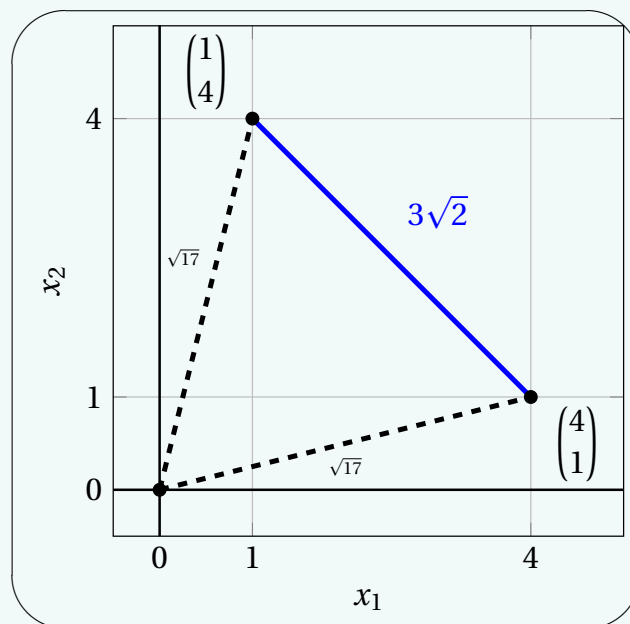
Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2,$$

The norm of the difference of the two vectors is

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{(1-4)^2 + (4-1)^2} = \sqrt{(-3)^2 + 3^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

Graphically, we have:



Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n,$$

the norm of the **sum** of the two vectors is:

$$\|\mathbf{p} + \mathbf{w}\|_2 = \sqrt{\sum_{j=1}^n (p_j + w_j)^2}$$

This quantity represents the **length of the diagonal** of the parallelogram spanned by the vectors $\mathbf{p}$ and $\mathbf{w}$, both originating from the origin.

Example 4: ℓ_2 norm of the sum of vectors

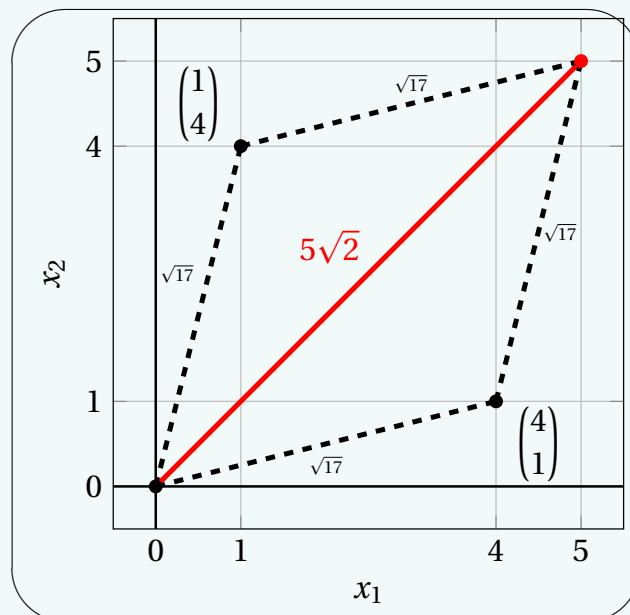
Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2,$$

The norm of the sum of the two vectors is:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{5^2 + 5^2} = 5\sqrt{2}$$

Graphically, we have:



3.1 Cauchy–Schwarz inequality

Observation 4: Cauchy–Schwarz inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$|\mathbf{p}' \mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \quad (6)$$

Proof. If $\mathbf{p} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$ (or both), the inequality clearly holds.

Assume $\mathbf{p} \neq \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$. Consider the vector:

$$\mathbf{u} = \alpha \mathbf{p} + \beta \mathbf{w}, \quad \text{with } \alpha, \beta \in \mathbb{R}$$

Then, for every $\alpha, \beta \in \mathbb{R}$, we have:

$$\begin{aligned} \underbrace{\mathbf{u}' \mathbf{u}}_{\geq 0} &= (\alpha \mathbf{p} + \beta \mathbf{w})' (\alpha \mathbf{p} + \beta \mathbf{w}) \\ &= \alpha^2 \mathbf{p}' \mathbf{p} + 2\alpha\beta \mathbf{p}' \mathbf{w} + \beta^2 \mathbf{w}' \mathbf{w} \geq 0 \end{aligned}$$

Now choose:

$$\alpha = \mathbf{w}' \mathbf{w}, \quad \beta = -\mathbf{p}' \mathbf{w}$$

Substituting, we get:

$$\begin{aligned} &(\mathbf{w}' \mathbf{w})^2 \mathbf{p}' \mathbf{p} - 2(\mathbf{p}' \mathbf{w})^2 (\mathbf{w}' \mathbf{w}) + (\mathbf{p}' \mathbf{w})^2 (\mathbf{w}' \mathbf{w}) \\ &= \mathbf{w}' \mathbf{w} ((\mathbf{w}' \mathbf{w}) \mathbf{p}' \mathbf{p} - (\mathbf{p}' \mathbf{w})^2) \geq 0 \end{aligned}$$

Since $\mathbf{w}' \mathbf{w} = \|\mathbf{w}\|_2^2 > 0$, dividing both sides yields:

$$(\mathbf{p}' \mathbf{w})^2 \leq \mathbf{p}' \mathbf{p} \mathbf{w}' \mathbf{w} \implies |\mathbf{p}' \mathbf{w}| \leq \sqrt{\mathbf{p}' \mathbf{p} \mathbf{w}' \mathbf{w}} = \sqrt{\mathbf{p}' \mathbf{p}} \sqrt{\mathbf{w}' \mathbf{w}} = \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$$

□

3.2 Properties

- The ℓ_2 norm satisfies the three fundamental properties that define a norm.

Observation 5: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_2 \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_2 = 0 \iff \mathbf{p} = \mathbf{0} \quad (7)$$

Proof. From the definition $\|\mathbf{p}\|_2 = \sqrt{\mathbf{p}'\mathbf{p}}$.

1. **Non-negativity:** Since $\mathbf{p}'\mathbf{p} = \sum_{j=1}^n p_j^2 \geq 0$ (as a sum of squares), and the square root function yields non-negative values, we have $\|\mathbf{p}\|_2 \geq 0$ for all $\mathbf{p} \in \mathbb{R}^n$.

2. **Definiteness:**

($\Rightarrow$) If $\|\mathbf{p}\|_2 = 0$, then $\sqrt{\mathbf{p}'\mathbf{p}} = 0$, hence $\mathbf{p}'\mathbf{p} = 0$, i.e.

$$\sum_{j=1}^n p_j^2 = 0.$$

Since each $p_j^2 \geq 0$, the sum equals zero only if $p_j^2 = 0$ for all $j \in \{1, 2, \dots, n\}$, hence $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, i.e., $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $\mathbf{p}'\mathbf{p} = 0$, so $\|\mathbf{p}\|_2 = \sqrt{0} = 0$.

□

Observation 6: Absolute homogeneity

For every column vector $\mathbf{p} \in \mathbb{R}^n$ and scalar $\lambda \in \mathbb{R}$, we have:

$$\|\lambda \mathbf{p}\|_2 = |\lambda| \|\mathbf{p}\|_2 \quad (8)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_2 = \sqrt{\sum_{j=1}^n (\lambda p_j)^2} = \sqrt{\sum_{j=1}^n \lambda^2 p_j^2} = |\lambda| \sqrt{\sum_{j=1}^n p_j^2} = |\lambda| \|\mathbf{p}\|_2$$

□

Observation 7: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_2 \leq \|\mathbf{p}\|_2 + \|\mathbf{w}\|_2 \quad (9)$$

Proof. For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} + \mathbf{w}\|_2^2 &= (\mathbf{p} + \mathbf{w})' (\mathbf{p} + \mathbf{w}) = \mathbf{p}' \mathbf{p} + 2 \mathbf{p}' \mathbf{w} + \mathbf{w}' \mathbf{w} = \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2 \mathbf{p}' \mathbf{w} \\ &\leq \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2 \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \quad (\text{by Cauchy-Schwarz inequality}) \\ &= (\|\mathbf{p}\|_2 + \|\mathbf{w}\|_2)^2 \end{aligned}$$

Accordingly, we obtain:

$$\|\mathbf{p} + \mathbf{w}\|_2^2 \leq (\|\mathbf{p}\|_2 + \|\mathbf{w}\|_2)^2 \implies \|\mathbf{p} + \mathbf{w}\|_2 \leq \|\mathbf{p}\|_2 + \|\mathbf{w}\|_2$$

□

- In $\mathbb{R}^2$, the triangle inequality states that **for any triangle the sum of the lengths of any two sides must be greater than or equal to the length of the remaining side.**

Example 5: triangle inequality

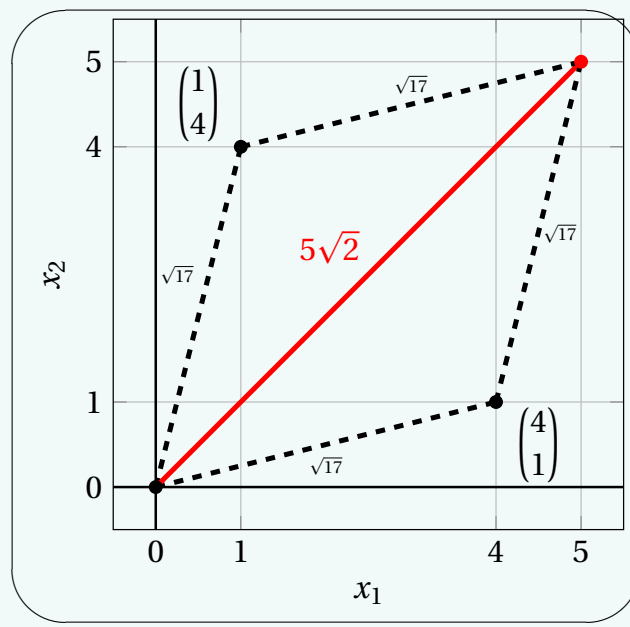
Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$, we have:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{4^2 + 1^2} = \sqrt{16 + 1} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{(1+4)^2 + (4+1)^2} = \sqrt{5^2 + 5^2} = 5\sqrt{2}$$

$$5\sqrt{2} \leq \sqrt{17} + \sqrt{17}$$



- Another form of the triangle inequality is:

$$\|\mathbf{r}\|_2 \leq \|\mathbf{r} - \mathbf{s}\|_2 + \|\mathbf{s}\|_2 \quad \text{that is} \quad \|\mathbf{r}\|_2 - \|\mathbf{s}\|_2 \leq \|\mathbf{r} - \mathbf{s}\|_2, \quad \forall \mathbf{r}, \mathbf{s} \in \mathbb{R}^n \quad (10)$$

To obtain it, it suffices to set in (9):

$$\mathbf{p} = \mathbf{r} - \mathbf{s}, \quad \mathbf{w} = \mathbf{s}$$

Observation 8: reverse triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$|\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2 \quad (11)$$

Proof. For every two vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} - \mathbf{w}\|_2^2 &= (\mathbf{p} - \mathbf{w})' (\mathbf{p} - \mathbf{w}) = \mathbf{p}' \mathbf{p} - 2 \mathbf{p}' \mathbf{w} + \mathbf{w}' \mathbf{w} = \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 - 2 \mathbf{p}' \mathbf{w} \\ &\geq \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 - 2 \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \quad (\text{by Cauchy-Schwarz inequality}) \\ &= (\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2)^2 \end{aligned}$$

Accordingly, we obtain:

$$(\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2)^2 \leq \|\mathbf{p} - \mathbf{w}\|_2^2 \implies |\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2$$

□

- In $\mathbb{R}^2$, the reverse triangle inequality states that **for any triangle the length of any side must be greater than or equal to the difference between the lengths of the other two sides.**

Example 6: reverse triangle inequality

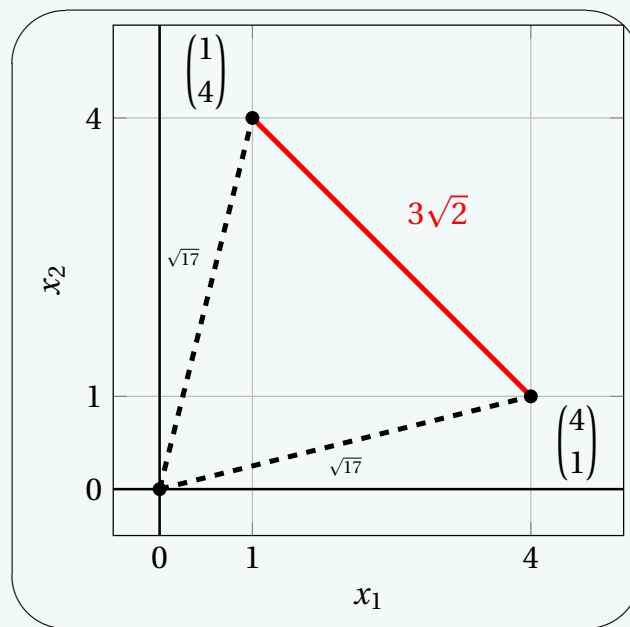
Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$, we have:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{4^2 + 1^2} = \sqrt{16 + 1} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{(1-4)^2 + (4-1)^2} = 3\sqrt{2}$$

$$|\sqrt{17} - \sqrt{17}| \leq 3\sqrt{2}$$



4 Generalized ℓ_2 norm

Definition 3: generalized ℓ_2 norm

Given a symmetric positive definite matrix

$$\underbrace{\begin{pmatrix} q_{11} & q_{12} & \dots & q_{1n} \\ q_{21} & q_{22} & \dots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{n1} & q_{n2} & \dots & q_{nn} \end{pmatrix}}_{\mathbf{Q}} \in \mathbb{R}^{n \times n}$$

the **generalized ℓ_2 norm** of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_{\mathbf{Q}} = \sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} p_i p_j} = \sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}} \quad (12)$$

- The **generalized ℓ_2 norm** extends the standard Euclidean norm by incorporating a positive definite matrix $\mathbf{Q}$, which weights and potentially correlates the components of the vector.
- When $\mathbf{Q}$ is the identity matrix $\mathbf{I}$, the generalized ℓ_2 norm reduces to the standard ℓ_2 norm.
- The matrix $\mathbf{Q}$ must be **positive definite** to ensure that $\mathbf{p}' \mathbf{Q} \mathbf{p} \geq 0$ for all $\mathbf{p} \in \mathbb{R}^n$, making the square root well-defined, and that $\mathbf{p}' \mathbf{Q} \mathbf{p} = 0$ only for $\mathbf{p} = \mathbf{0}$.
- In two dimensions, all the points at generalized ℓ_2 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\}$$

are the points that lie on or inside an **ellipse** centered at the origin, whose size, shape, and rotation are determined by the matrix $\mathbf{Q}$. In particular, the eigenvectors of $\mathbf{Q}$ determine the principal axes and the orientation, while the eigenvalues determine the lengths of the semi-axes.

Example 7: generalized ℓ_2 norm

In $\mathbb{R}^2$, we consider the following diagonal positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

Since $\mathbf{Q}$ is **diagonal**, the eigenvalues are the diagonal entries:

$$\lambda_1 = 4, \quad \lambda_2 = 1$$

and the eigenvectors are the standard basis vectors:

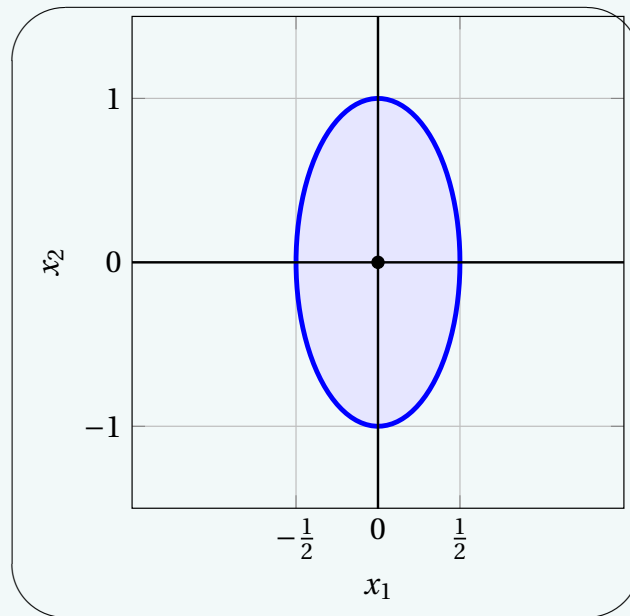
$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{for } \lambda_1 = 4), \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_2 = 1)$$

Semi-axes lengths and orientation:

- The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{1}} = 1$.
- Since the eigenvectors are aligned with the coordinate axes, the ellipse is **not rotated**.
- The principal axes coincide with the x_1 and x_2 axes.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \sqrt{4x_1^2 + x_2^2} \leq 1 \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \frac{x_1^2}{(1/2)^2} + \frac{x_2^2}{1^2} \leq 1 \right\}$$



Example 8: generalized ℓ_2 norm

In $\mathbb{R}^2$, we consider the following diagonal positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} \frac{1}{9} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$$

Since $\mathbf{Q}$ is **diagonal**, the eigenvalues are the diagonal entries:

$$\lambda_1 = \frac{1}{9}, \quad \lambda_2 = \frac{1}{4}$$

and the eigenvectors are the standard basis vectors:

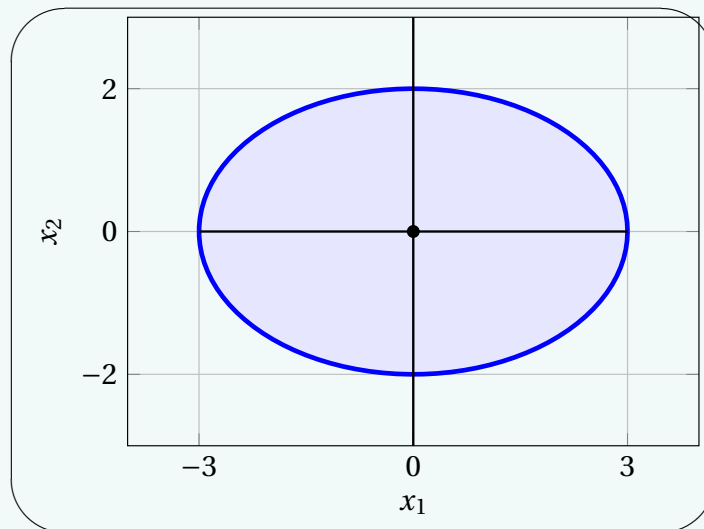
$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{for } \lambda_1 = \frac{1}{9}), \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_2 = \frac{1}{4})$$

Semi-axes lengths and orientation:

- The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{1/9}} = 3$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{1/4}} = 2$.
- Since the eigenvectors are aligned with the coordinate axes, the ellipse is **not rotated**.
- The principal axes coincide with the x_1 and x_2 axes.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \sqrt{\frac{x_1^2}{9} + \frac{x_2^2}{4}} \leq 1 \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \frac{x_1^2}{3^2} + \frac{x_2^2}{2^2} \leq 1 \right\}$$



Example 9: generalized ℓ_2 norm

In $\mathbb{R}^2$, we consider the following symmetric positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

The eigenvalues of $\mathbf{Q}$ are found by solving the characteristic equation:

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & -1 \\ -1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = 0$$

which gives:

$$\lambda_1 = 1, \quad \lambda_2 = 3$$

The corresponding eigenvectors are:

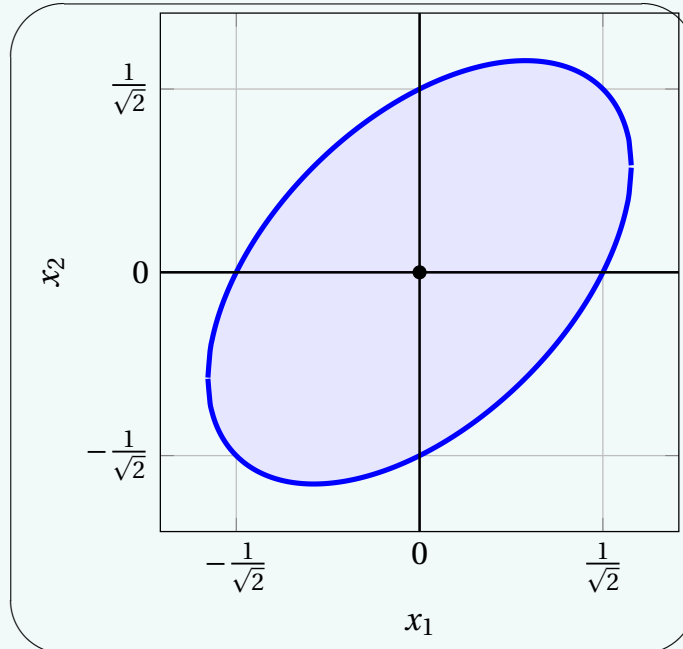
$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_1 = 1), \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (\text{for } \lambda_2 = 3)$$

Semi-axes lengths and orientation:

- The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{1}} = 1$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{3}}$.
- Since the eigenvectors are **not aligned** with the coordinate axes, the ellipse is **rotated**.
- The principal axes coincide with the directions of the eigenvectors.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \sqrt{2x_1^2 - 2x_1x_2 + 2x_2^2} \leq 1 \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : 2x_1^2 - 2x_1x_2 + 2x_2^2 \leq 1 \right\}$$



4.1 Generalized Cauchy–Schwarz inequality

Observation 9: generalized Cauchy–Schwarz inequality

For every symmetric positive definite matrix $Q \in \mathbb{R}^{n \times n}$ and pair of column vectors $p, w \in \mathbb{R}^n$, we have:

$$|p' Q w| \leq \|p\|_Q \|w\|_Q \quad (13)$$

Proof. If $p = 0$ or $w = 0$ (or both), the inequality clearly holds.

Assume $p \neq 0$ and $w \neq 0$. Consider the vector:

$$u = \alpha p + \beta w, \quad \text{with } \alpha, \beta \in \mathbb{R}$$

Then, for every $\alpha, \beta \in \mathbb{R}$, we have:

$$\begin{aligned} \underbrace{u' Q u}_{\geq 0} &= (\alpha p + \beta w)' Q (\alpha p + \beta w) \\ &= \alpha^2 p' Q p + 2\alpha\beta p' Q w + \beta^2 w' Q w \geq 0 \end{aligned}$$

where we used $w' Q p = p' Q w$, which holds since Q is symmetric.

Now choose:

$$\alpha = w' Q w, \quad \beta = -p' Q w$$

Substituting, we get:

$$\begin{aligned} &(w' Q w)^2 p' Q p - 2(p' Q w)^2 (w' Q w) + (p' Q w)^2 (w' Q w) \\ &= w' Q w ((w' Q w) p' Q p - (p' Q w)^2) \geq 0 \end{aligned}$$

Since $w' Q w = \|w\|_Q^2 > 0$, dividing both sides yields:

$$(p' Q w)^2 \leq p' Q p \cdot w' Q w \implies |p' Q w| \leq \sqrt{p' Q p} \sqrt{w' Q w} = \|p\|_Q \|w\|_Q$$

□

4.2 Properties

- The generalized ℓ_2 norm satisfies the three fundamental properties that define a norm.

Observation 10: Non-negativity and definiteness

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_{\mathbf{Q}} \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_{\mathbf{Q}} = 0 \iff \mathbf{p} = \mathbf{0} \quad (14)$$

Proof. By definition, $\|\mathbf{p}\|_{\mathbf{Q}} = \sqrt{\mathbf{p}'\mathbf{Q}\mathbf{p}}$. Since $\mathbf{Q}$ is positive definite, we have $\mathbf{p}'\mathbf{Q}\mathbf{p} > 0$ for all $\mathbf{p} \neq \mathbf{0}$, and $\mathbf{0}'\mathbf{Q}\mathbf{0} = 0$.

1. **Non-negativity:** For all $\mathbf{p} \in \mathbb{R}^n$, we have $\mathbf{p}'\mathbf{Q}\mathbf{p} \geq 0$. Therefore

$$\|\mathbf{p}\|_{\mathbf{Q}} = \sqrt{\mathbf{p}'\mathbf{Q}\mathbf{p}} \geq 0.$$

2. **Definiteness:**

($\Rightarrow$) If $\|\mathbf{p}\|_{\mathbf{Q}} = 0$, then

$$\sqrt{\mathbf{p}'\mathbf{Q}\mathbf{p}} = 0 \implies \mathbf{p}'\mathbf{Q}\mathbf{p} = 0,$$

which implies $\mathbf{p} = \mathbf{0}$ (by positive definiteness of $\mathbf{Q}$).

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then

$$\mathbf{p}'\mathbf{Q}\mathbf{p} = \mathbf{0}'\mathbf{Q}\mathbf{0} = 0,$$

and therefore $\|\mathbf{p}\|_{\mathbf{Q}} = \sqrt{0} = 0$.

□

Observation 11: Absolute homogeneity

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$, scalar $\lambda \in \mathbb{R}$, and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_{\mathbf{Q}} = |\lambda| \|\mathbf{p}\|_{\mathbf{Q}} \quad (15)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\lambda \mathbf{p}\|_{\mathbf{Q}} &= \sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} \lambda p_i \lambda p_j} \\ &= \sqrt{\sum_{i=1}^n \sum_{j=1}^n \lambda^2 q_{ij} p_i p_j} \\ &= |\lambda| \sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} p_i p_j} \\ &= |\lambda| \|\mathbf{p}\|_{\mathbf{Q}} \end{aligned}$$

□

Observation 12: Triangle inequality

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and pairs of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_{\mathbf{Q}} \leq \|\mathbf{p}\|_{\mathbf{Q}} + \|\mathbf{w}\|_{\mathbf{Q}} \quad (16)$$

Proof. For every $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} + \mathbf{w}\|_{\mathbf{Q}}^2 &= (\mathbf{p} + \mathbf{w})' \mathbf{Q} (\mathbf{p} + \mathbf{w}) = \mathbf{p}' \mathbf{Q} \mathbf{p} + 2 \mathbf{p}' \mathbf{Q} \mathbf{w} + \mathbf{w}' \mathbf{Q} \mathbf{w} \\ &= \|\mathbf{p}\|_{\mathbf{Q}}^2 + \|\mathbf{w}\|_{\mathbf{Q}}^2 + 2 \mathbf{p}' \mathbf{Q} \mathbf{w} \\ &\leq \|\mathbf{p}\|_{\mathbf{Q}}^2 + \|\mathbf{w}\|_{\mathbf{Q}}^2 + 2 \|\mathbf{p}\|_{\mathbf{Q}} \|\mathbf{w}\|_{\mathbf{Q}} \quad (\text{by gen. Cauchy-Schwarz inequality}) \\ &= (\|\mathbf{p}\|_{\mathbf{Q}} + \|\mathbf{w}\|_{\mathbf{Q}})^2 \end{aligned}$$

Accordingly, we obtain:

$$\|\mathbf{p} + \mathbf{w}\|_{\mathbf{Q}}^2 \leq (\|\mathbf{p}\|_{\mathbf{Q}} + \|\mathbf{w}\|_{\mathbf{Q}})^2 \implies \|\mathbf{p} + \mathbf{w}\|_{\mathbf{Q}} \leq \|\mathbf{p}\|_{\mathbf{Q}} + \|\mathbf{w}\|_{\mathbf{Q}}$$

□

5 ℓ_∞ norm

Definition 4: ℓ_∞ norm

The ℓ_∞ **norm** (also called **maximum norm**, **supremum norm**, or **Chebyshev norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_p \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|p\|_\infty = \max\{|p_j| : j \in \{1, 2, \dots, n\}\} \quad (17)$$

- The ℓ_∞ **norm** represents the maximum absolute value among all components.
- The ℓ_∞ norm is half the side length of the smallest hypercube (aligned with the axes and centered at the origin) that contains the vector.

Example 10: ℓ_∞ norm

Consider the vectors:

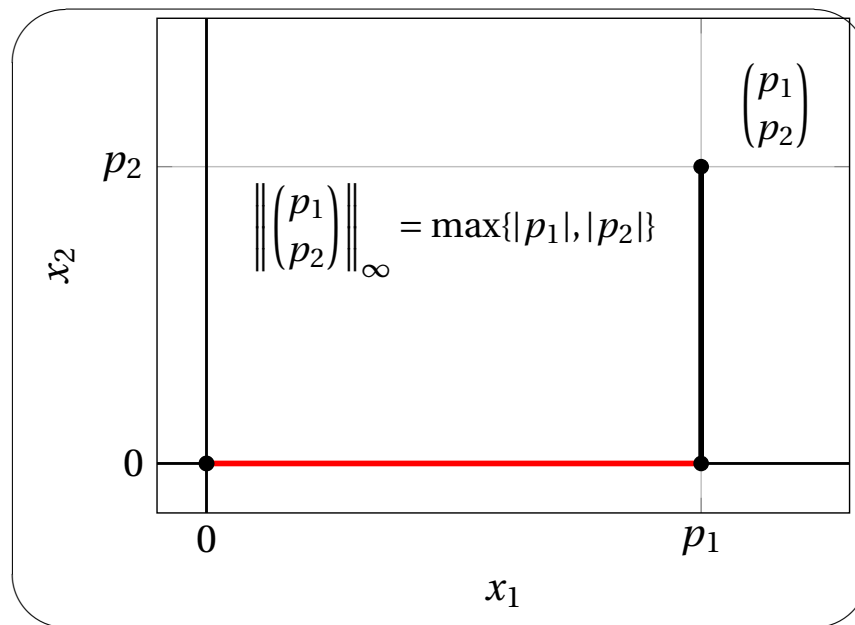
$$p = \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} \in \mathbb{R}^3, \quad w = \begin{pmatrix} -1 \\ 5 \\ -5 \\ 2 \end{pmatrix} \in \mathbb{R}^4$$

Their ℓ_∞ norms are:

$$\left\| \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} \right\|_\infty = \max\{|3|, |-7|, |2|\} = \max\{3, 7, 2\} = 7,$$

$$\left\| \begin{pmatrix} -1 \\ 5 \\ -5 \\ 2 \end{pmatrix} \right\|_\infty = \max\{|-1|, |5|, |-5|, |2|\} = \max\{1, 5, 5, 2\} = 5$$

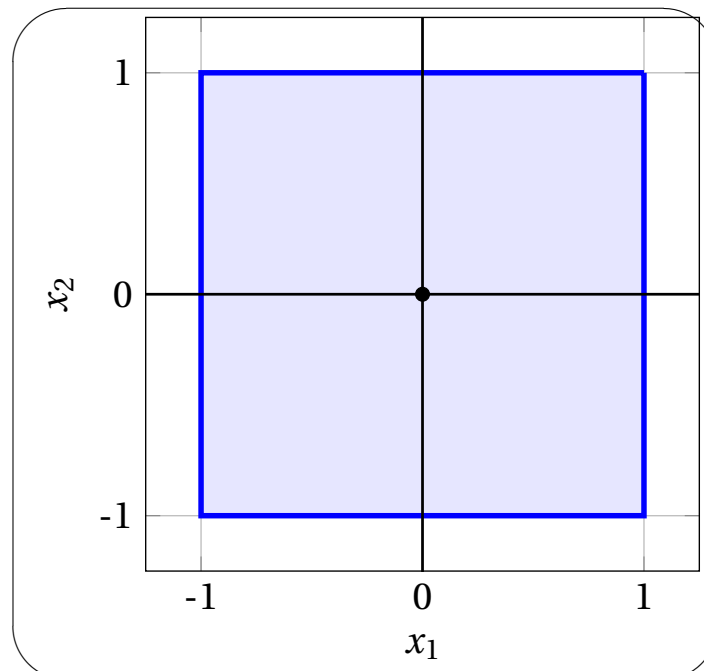
- The ℓ_∞ **norm** of a column vector in $\mathbb{R}^n$ represents the **maximum absolute value of its components**. In $\mathbb{R}^2$, this corresponds to the **Chebyshev distance** (the maximum of the absolute differences in coordinates), as illustrated by the following picture:



- In two dimensions, all the points at ℓ_∞ norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_\infty \leq 1\}$$

are the points that lie on or inside a **square** (axis-aligned) centered at the origin.



5.1 Properties

- The ℓ_∞ norm satisfies the three fundamental properties that define a norm.

Observation 13: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_\infty \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_\infty = 0 \iff \mathbf{p} = \mathbf{0} \quad (18)$$

Proof. From the definition $\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, 2, \dots, n\}\}$.

1. **Non-negativity:** Since $|p_j| \geq 0$ for all $j \in \{1, 2, \dots, n\}$, and the maximum of non-negative numbers is non-negative, we have

$$\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, 2, \dots, n\}\} \geq 0 \quad \text{for all } \mathbf{p} \in \mathbb{R}^n.$$

2. **Definiteness:**

($\Rightarrow$) If $\|\mathbf{p}\|_\infty = 0$, then

$$\max\{|p_j| : j \in \{1, 2, \dots, n\}\} = 0.$$

Since

$$|p_j| \leq \max\{|p_k| : k \in \{1, 2, \dots, n\}\} \quad \text{for all } j \in \{1, 2, \dots, n\},$$

it follows that

$$|p_j| \leq 0 \quad \text{for all } j \in \{1, 2, \dots, n\}.$$

But $|p_j| \geq 0$ always holds, hence $|p_j| = 0$ for all j , i.e. $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, and therefore $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $p_j = 0$ for all $j \in \{1, 2, \dots, n\}$, so

$$\|\mathbf{p}\|_\infty = \max\{|0| : j \in \{1, 2, \dots, n\}\} = 0.$$

□

Observation 14: Absolute homogeneity

For every scalar $\lambda \in \mathbb{R}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_\infty = |\lambda| \|\mathbf{p}\|_\infty \quad (19)$$

Proof. For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\lambda \mathbf{p}\|_\infty &= \max\{|\lambda p_j| : j \in \{1, 2, \dots, n\}\} \\ &= \max\{|\lambda| |p_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{property of absolute value}) \\ &= |\lambda| \max\{|p_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{factor constant out of max}) \\ &= |\lambda| \|\mathbf{p}\|_\infty \end{aligned}$$

□

Observation 15: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_\infty \leq \|\mathbf{p}\|_\infty + \|\mathbf{w}\|_\infty \quad (20)$$

Proof. For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned} \|\mathbf{p} + \mathbf{w}\|_\infty &= \max\{|p_j + w_j| : j \in \{1, 2, \dots, n\}\} \\ &\leq \max\{|p_j| + |w_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{triangle inequality for absolute value}) \\ &\leq \max\{|p_j| : j \in \{1, 2, \dots, n\}\} + \max\{|w_j| : j \in \{1, 2, \dots, n\}\} \quad (\text{max distribution}) \\ &= \|\mathbf{p}\|_\infty + \|\mathbf{w}\|_\infty \end{aligned}$$

where in the second inequality we used the fact that

$$|p_j| \leq \max\{|p_k| : k \in \{1, 2, \dots, n\}\}$$

and

$$|w_j| \leq \max\{|w_k| : k \in \{1, 2, \dots, n\}\} \quad \text{for all } j \in \{1, 2, \dots, n\}.$$

□

6 Comparison of the ℓ_1 , ℓ_2 and ℓ_∞ norms

- The ℓ_1 , ℓ_2 and ℓ_∞ norms of the same vector are, in general, different numbers. However, they are related by simple inequalities.

Observation 16: Inequalities between norms

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_\infty \quad (21)$$

and

$$\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2 \quad (22)$$

Proof. Let $k \in \{1, 2, \dots, n\}$ be an index such that $|p_k| = \|\mathbf{p}\|_\infty$.

1. $\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2$: since all the terms p_j^2 are non-negative, we have

$$\|\mathbf{p}\|_\infty^2 = p_k^2 \leq \sum_{j=1}^n p_j^2 = \|\mathbf{p}\|_2^2$$

and, taking the square root of both (non-negative) sides, $\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2$.

2. $\|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1$: expanding the square of the sum, we have

$$\|\mathbf{p}\|_1^2 = \left(\sum_{j=1}^n |p_j| \right)^2 = \sum_{j=1}^n |p_j|^2 + \underbrace{\sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n |p_i| |p_j|}_{\geq 0} \geq \sum_{j=1}^n p_j^2 = \|\mathbf{p}\|_2^2$$

and, taking the square root of both (non-negative) sides, $\|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1$.

3. $\|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_\infty$: since $|p_j| \leq \|\mathbf{p}\|_\infty$ for all $j \in \{1, 2, \dots, n\}$, we have

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j| \leq \sum_{j=1}^n \|\mathbf{p}\|_\infty = n \|\mathbf{p}\|_\infty$$

4. $\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2$: consider the column vectors

$$\mathbf{a} = \begin{pmatrix} |p_1| \\ |p_2| \\ \vdots \\ |p_n| \end{pmatrix} \in \mathbb{R}^n \quad \text{and} \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^n$$

for which $\mathbf{1}' \mathbf{a} = \|\mathbf{p}\|_1$, $\|\mathbf{a}\|_2 = \|\mathbf{p}\|_2$ and $\|\mathbf{1}\|_2 = \sqrt{n}$. By the Cauchy–Schwarz inequality (6), we get

$$\|\mathbf{p}\|_1 = \mathbf{1}' \mathbf{a} \leq \|\mathbf{1}\|_2 \|\mathbf{a}\|_2 = \sqrt{n} \|\mathbf{p}\|_2$$

□

Example 11: inequalities between norms

Consider the vector:

$$\mathbf{p} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \in \mathbb{R}^3$$

Its norms are:

$$\|\mathbf{p}\|_\infty = \max\{1, 2, 2\} = 2, \quad \|\mathbf{p}\|_2 = \sqrt{1+4+4} = 3, \quad \|\mathbf{p}\|_1 = 1+2+2 = 5$$

and, with $n = 3$, we have:

$$\underbrace{2}_{\|p\|_\infty} \leq \underbrace{3}_{\|p\|_2} \leq \underbrace{5}_{\|p\|_1} \leq \underbrace{6}_{3\|p\|_\infty} \quad \text{and} \quad \underbrace{5}_{\|p\|_1} \leq \underbrace{3\sqrt{3}}_{\sqrt{3}\|p\|_2} \approx 5.196$$

- In two dimensions, the inequalities $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ mean that the three unit balls are nested: the ℓ_1 diamond lies inside the ℓ_2 unit circle, which in turn lies inside the ℓ_∞ square.

